

Ghazal Mousavi

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 LinkedIn

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 Scholar

Personal Profile

A self-motivated and passionate Electrical Engineering graduate from the University of Tehran. Interested in pursuing a M.Sc. degree in Telecommunications and Signal Processing. Experienced in decentralized massive MIMO detection and contributing to a research paper, with interests in signal processing, wireless communications, and coding theory.

Education

University of Tehran: B.Sc. Electrical Engineering Sept 2021 – March 2026

- GPA: 18.35 / 20, ranked 8th out of 101 students

University of Tehran: Minor Computer Engineering Sept 2023 – March 2026

- **Relevant Coursework:** Advanced Programming, Discrete Mathematics, Artificial Intelligence, Data Structures and Algorithms

Publications

Newton-Based Mini-Batch Detection in Decentralized Large-Scale Antenna Systems (Accepted at [ICEE 2026](#)) November 2025

Ghazal Mousavi, Mojtaba Amiri, Ali Olfat

Machine Learning Evaluation Metric Discrepancies Across Programming Languages and Their Components in Medical Imaging Domains: Need for Standardization March 2025

Mohammad R. Salmanpour, Morteza Alizadeh, *Ghazal Mousavi*, Saba Sadeghi, Sajad Amiri, Mehrdad Oveisi, Arman Rahmim, Ilker Hacihaliloglu

[10.1109/ACCESS.2025.3549702](https://doi.org/10.1109/ACCESS.2025.3549702)

AllMetrics: A Unified Python Library for Standardized Metric Evaluation in Machine Learning July 2025

Morteza Alizadeh, Mehrdad Oveisi, Sonya Falahati, *Ghazal Mousavi*

[10.1109/NSS/MIC/RTSD57106.2025.11286722](https://doi.org/10.1109/NSS/MIC/RTSD57106.2025.11286722)

Research Experiences

Research Assistant: University of Tehran March 2025 – Present

Signal Processing & Communication Systems Laboratory, supervised by [Prof. Ali Olfat](#)

- Working on integrating machine learning with Markov Chain Monte Carlo (MCMC) methods for decentralized massive MIMO detection.

Researcher: [VirCollab](#), Canada – Remote June 2024 – June 2025

- Worked with various machine learning evaluation metrics and evaluated model performance across different metrics.
- Contributed to an image-to-image translation project, gaining hands-on experience with generator and discriminator models for ultrasound-to-MRI conversion.

Research Assistant: [Monash University](#), Australia – Remote July 2024 – January 2025

Under the supervision of [Prof. Mehrtash Harandi](#)

- Working on enhancing machine unlearning through vision transformers for image classification.

Teaching Assistant Experiences

- **Engineering Mathematics**
Instructor: [Dr. Mahdi Tale Masouleh](#)
- **Signals and Systems**
Instructor: [Dr. Saeed Akhavan](#)
- **Principles of Communication Systems**
Instructor: [Prof. Ali Olfat](#)
- **Communication Systems II**
Instructor: [Prof. Ali Olfat](#)
- **Statistics & Probability**
Instructor: [Dr. Amir Masoud Rabiei](#)
- **Principles of Communication Systems**
Instructor: [Hamideh Zamanpour](#)

Projects

Wireless Communication Projects

- Implemented OFDM transmission of 10^7 bits using QPSK modulation over an AWGN channel
- Simulated time diversity over Rayleigh fading channels
- Tools used: MATLAB

Deep Learning and Model Fine-Tuning Projects

- Fine-tuning large language models for Persian conversation using LLaMA 3.2-3B
- Triplet Variational Autoencoder (Tri-VAE) for tumor detection in MRI scans
- Vision Transformer (ViT) for white blood cell classification

Blind Source Separation (BSS) and EEG Analysis Projects

- Implemented Common Spatial Pattern (CSP) and Canonical Correlation Analysis (CCA) techniques for Brain-Computer Interface (BCI) problems to extract optimal EEG features
- Solved Blind Source Separation problems using methods like maximizing kurtosis and minimizing D-Kullback divergence
- Addressed sparse recovery problems, non-stationary and uncorrelated sources, and mutually uncorrelated sources
- Tools used: MATLAB

Skills

Programming Languages: C++, Python, R, Verilog

Tools & Frameworks: MATLAB, TensorFlow, PyTorch, Git, Ansys HFSS, $\text{\LaTeX}$

IELTS: Overall Band Score **7.0** (Listening: 7.5, Reading: 7.0, Writing: 6.5, Speaking: 7.0) *[Advanced Level]*